

ECE 3337: Signals & Systems Analysis

Lecture 19: Fourier Analysis Techniques

**Reference: Chapter 5
(Section 5-3 – 5-4)**

Please turn off your gadgets now



Recap: Fourier Series

We discussed Fourier's famous claim that a general periodic signal

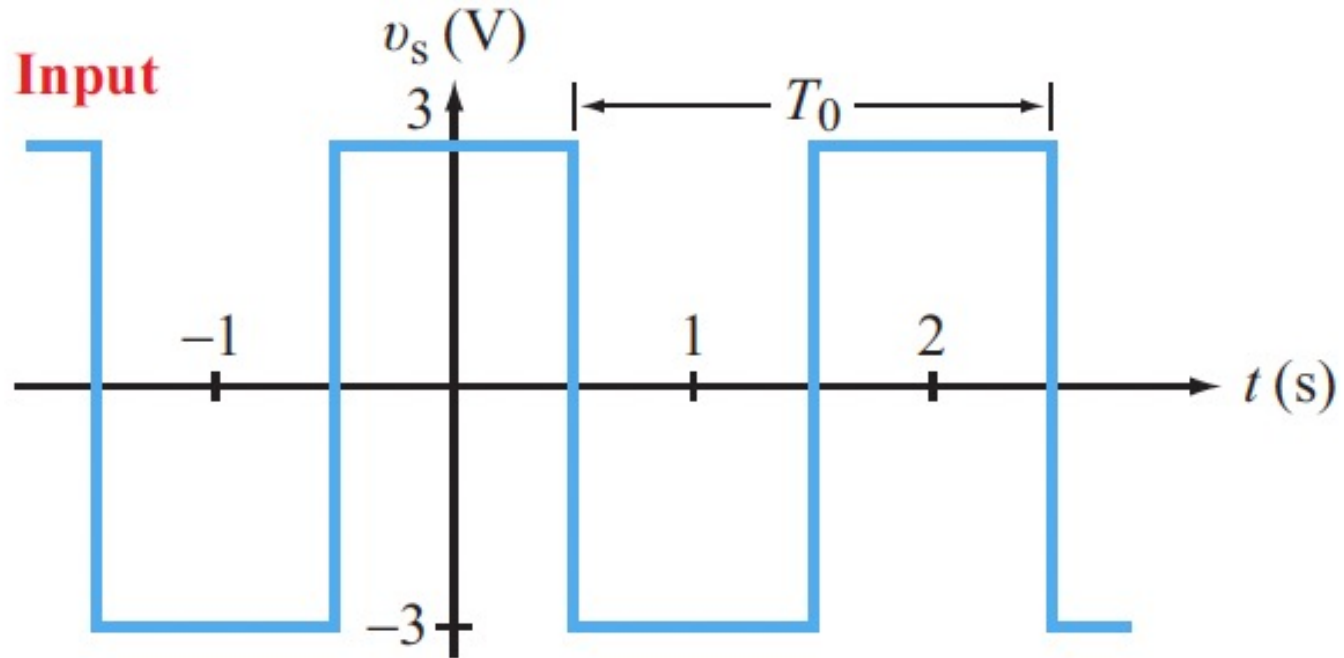
$$x(t) = x(t + nT_0)$$

can be expressed as a linear combination of sinusoids, as follows:

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$



Recap: Square Wave Example



FOURIER SERIES REPRESENTATION

$$v_s(t) = \frac{12}{\pi} \cos \omega_0 t - \frac{12}{3\pi} \cos 3\omega_0 t + \frac{12}{5\pi} \cos 5\omega_0 t - \dots$$

$$\omega_0 = \frac{2\pi}{T_0}$$

LET US DERIVE THIS FORMULA TODAY !



Recap: Trigonometric Identities

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y \text{ (basic formula 1)}$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y \text{ (basic formula 2)}$$

$$\Rightarrow \sin(2x) = 2 \sin x \cos x$$

$$\Rightarrow \cos(2x) = \cos^2 x - \sin^2 x$$

$$\Rightarrow \sin x \sin y = \frac{1}{2} [\cos(x - y) - \cos(x + y)]$$

$$\Rightarrow \cos x \cos y = \frac{1}{2} [\cos(x - y) + \cos(x + y)]$$

ETC.



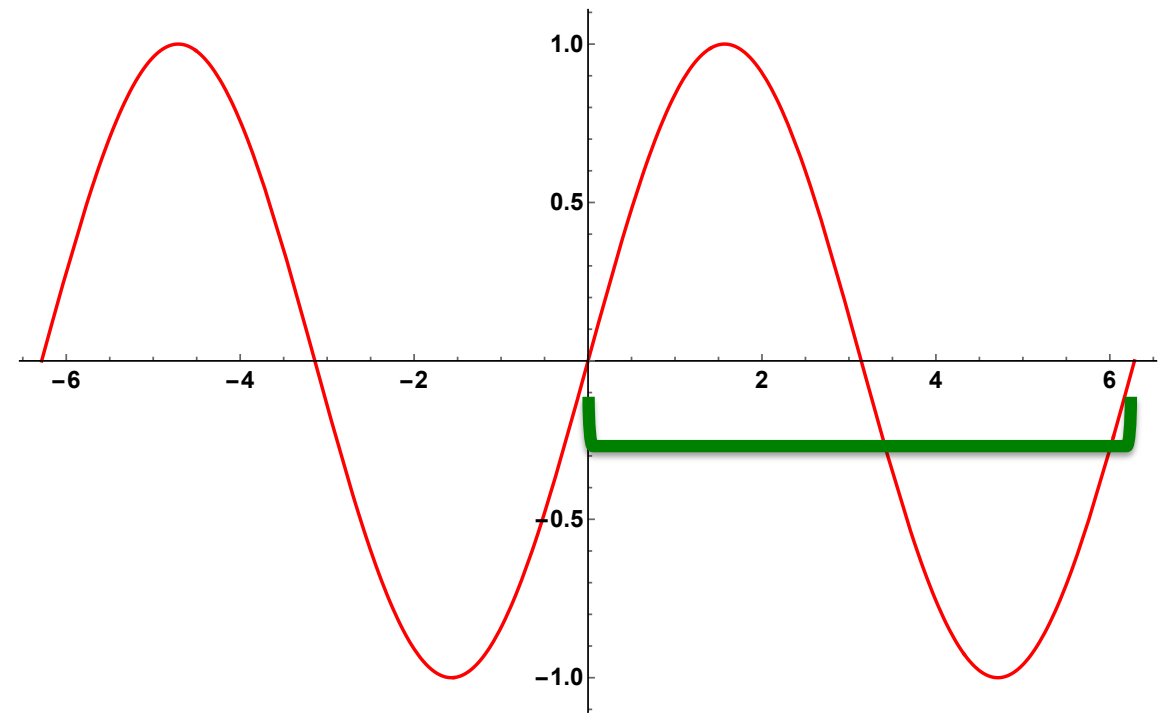
Integrals of Sinusoids

The integral of a sinusoid over one time period is zero

$$\int_0^{T_0} \sin(\omega_0 t) dt = 0$$

By extension, this is also true over multiple periods:

$$\int_0^{T_0} \sin(n\omega_0 t) dt = 0$$





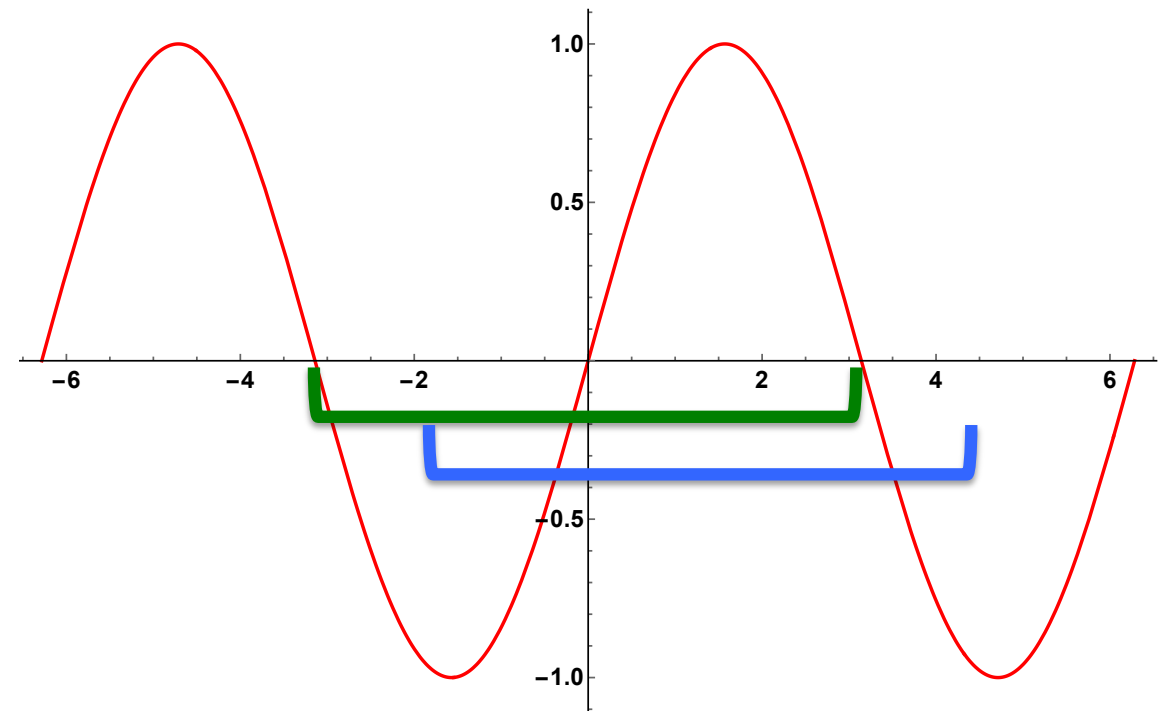
Integrals of Sinusoids

Introducing a phase shift does not change the answer, since it merely re-arranges the same set of points

$$\int_0^{T_0} \sin(n\omega_0 t + \phi) dt = 0$$

Similarly,

$$\int_0^{T_0} \cos(n\omega_0 t + \phi) dt = 0$$





Putting this to Use

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

Integrate both sides over one period:

$$\int_0^{T_0} x(t) dt = \int_0^{T_0} a_0 dt + \underbrace{\int_0^{T_0} \left(\sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right) dt}_{\text{THIS WHOLE THING IS ZERO}}$$

$$\int_0^{T_0} x(t) dt = a_0 T_0 \Rightarrow a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

AVERAGE



The DC Offset

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

THIS IS JUST THE AVERAGE VALUE OF THE SIGNAL OVER A PERIOD

**REPRESENTS THE CONSTANT (NON-VARYING) TERM
IT IS CALLED THE “DC OFFSET”**



The Higher Coefficients

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

We next need a strategy for calculating

$$\{a_n, b_n\}, \quad n = 1, 2, 3, \dots$$

For this, we exploit some more math results



Products of Sinusoids

Recall (Appendix C) that:

$$\sin x \sin y = \frac{1}{2} [\cos(x - y) - \cos(x + y)]$$

$$\sin(n\omega_0 t) \sin(m\omega_0 t) = \frac{1}{2} [\cos((n - m)\omega_0 t) - \cos((n + m)\omega_0 t)]$$

$$\text{When } n = m, \quad \cos((n - m)\omega_0 t) = \cos(0) = 1$$

$$\Rightarrow \sin(n\omega_0 t) \sin(n\omega_0 t) = \frac{1}{2} [1 - \cos(2n\omega_0 t)]$$



Integrals of Products of Sinusoids

$$\begin{aligned} & \int_0^{T_0} \sin(n\omega_0 t) \sin(m\omega_0 t) dt \\ &= \int_0^{T_0} [\cos((n-m)\omega_0 t) - \cos((n+m)\omega_0 t)] dt \\ &= 0 \quad \text{when } n \neq m \end{aligned}$$



Integrals of Products of Sinusoids

But when $n = m$,

$$\sin(n\omega_0 t)\sin(m\omega_0 t) = \frac{1}{2}[1 - \cos(2n\omega_0 t)]$$

$$\Rightarrow \int_0^{T_0} \sin(n\omega_0 t)\sin(m\omega_0 t) dt = \frac{T_0}{2}$$

THIS PROPERTY HELPS US CALCULATE FOURIER COEFFICIENTS!



Summary of Integrals we will Need

Property	Integral
1	$\int_0^{T_0} \sin(n\omega_0 t + \phi) dt = 0$
2	$\int_0^{T_0} \cos(n\omega_0 t + \phi) dt = 0$
3	$\int_0^{T_0} \sin(n\omega_0 t + \phi_1) \sin(m\omega_0 t + \phi_2) dt = 0$ $n \neq m$
4	$\int_0^{T_0} \cos(n\omega_0 t + \phi_1) \cos(m\omega_0 t + \phi_2) dt = 0$ $n \neq m$
5	$\int_0^{T_0} \sin(n\omega_0 t + \phi_1) \cos(m\omega_0 t + \phi_2) dt = 0$
6	$\int_0^{T_0} \sin^2(n\omega_0 t + \phi) dt = T_0/2$
7	$\int_0^{T_0} \cos^2(n\omega_0 t + \phi) dt = T_0/2$



Calculating Fourier Coefficients

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$

LET'S MULTIPLY BY $\cos(m\omega_0 t)$ THROUGHOUT, AND INTEGRATE OVER ONE PERIOD

$$\begin{aligned} \int_0^{T_0} x(t) \cos(m\omega_0 t) dt &= \int_0^{T_0} a_0 \cos(m\omega_0 t) dt \\ &+ \int_0^{T_0} \cos(m\omega_0 t) \left(\sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right) dt \end{aligned}$$

THESE INTEGRALS ARE ZERO, EXCEPT WHEN $m = n$



LET'S CALCULATE THE TERM FOR $m = n$

$$\int_0^{T_0} x(t) \cos(n\omega_0 t) dt$$

$$= \int_0^{T_0} \cos(n\omega_0 t) (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)) dt$$

$$= \int_0^{T_0} \cos(n\omega_0 t) (a_n \cos(n\omega_0 t)) dt + \int_0^{T_0} \cos(n\omega_0 t) (b_n \sin(n\omega_0 t)) dt$$

$$= a_n \int_0^{T_0} \cos^2(n\omega_0 t) dt + 0 = a_n \frac{T_0}{2}$$

$$\Rightarrow a_n = \frac{2}{T_0} \int_0^{T_0} x(t) \cos(n\omega_0 t) dt$$



Calculating Fourier Coefficients

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

WE CAN SIMILARLY SHOW THAT:

$$b_n = \frac{2}{T_0} \int_0^{T_0} x(t) \sin(n\omega_0 t) dt$$

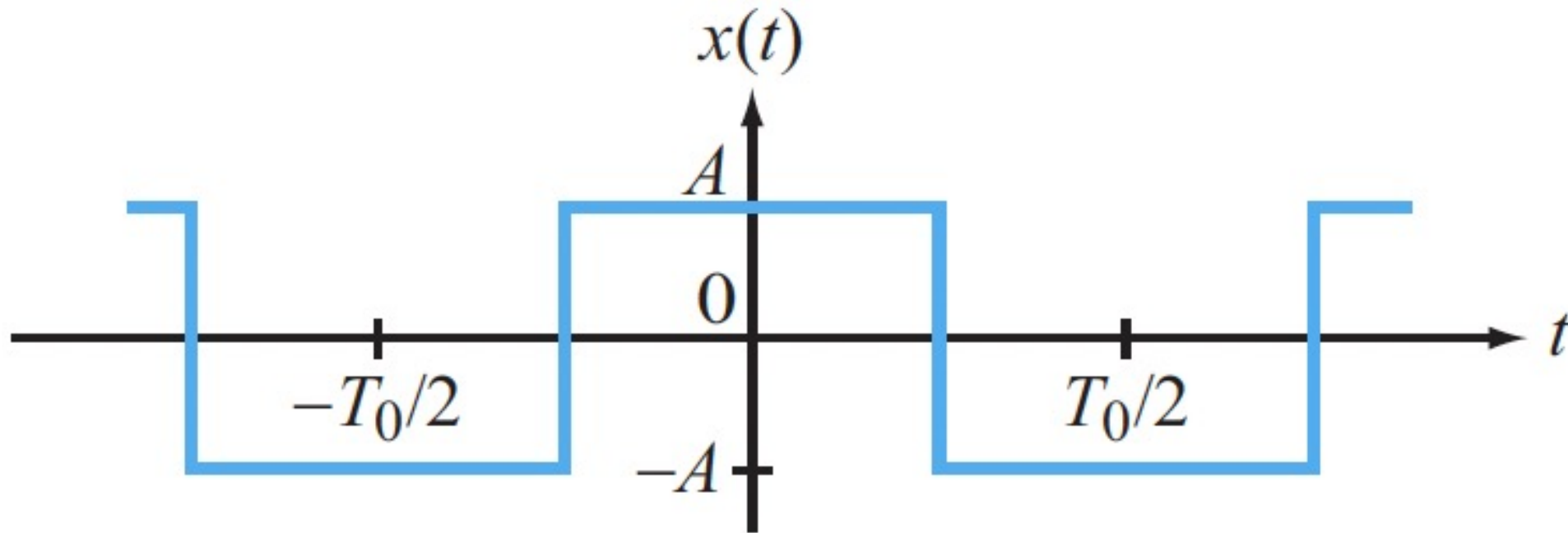
THIS IS IT!

**FOURIER JUST USED COLLEGE-LEVEL TRIGONOMETRY AND
CALCULUS TO DERIVE HIS REVOLUTIONARY THEOREM !**



Example

Let's calculate the Fourier series for the basic square wave:

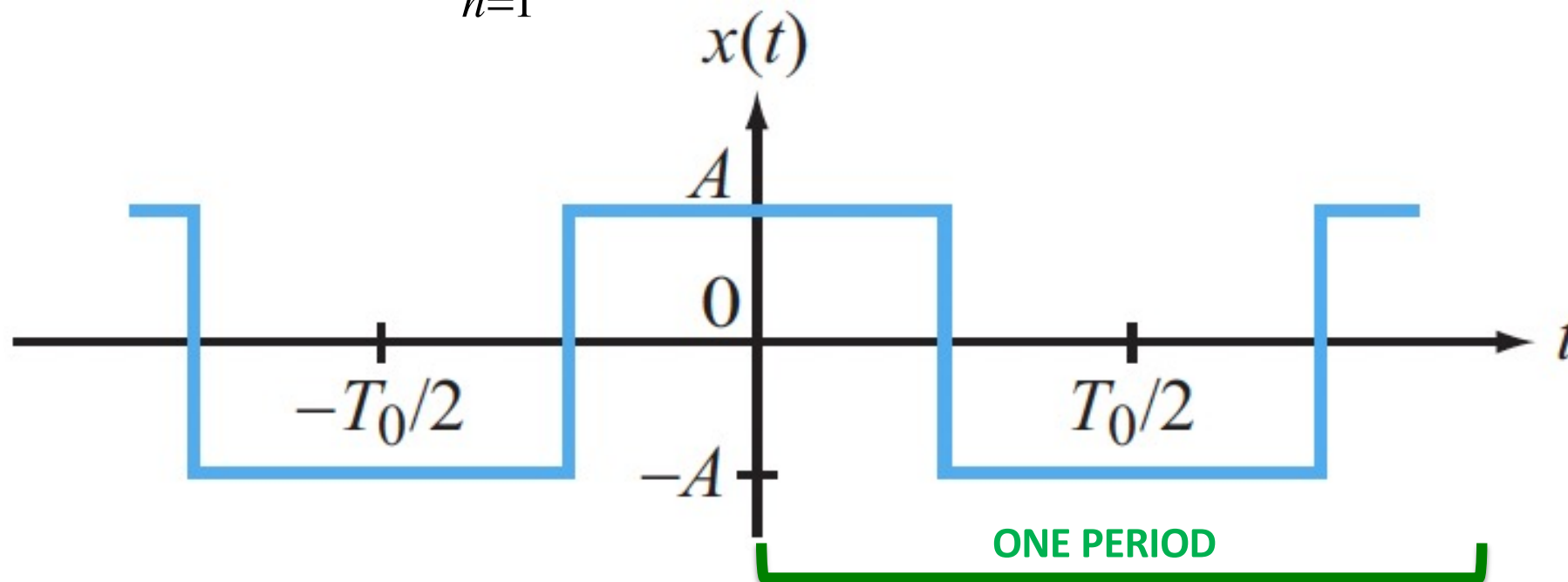


$$x(t) = \begin{cases} A & 0 < t < T_0 / 4 \\ -A & T_0 / 4 < t < 3T_0 / 4 \\ A & 3T_0 / 4 < t < T_0 \end{cases}$$



Example

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$



By looking at the waveform, can you guess anything about a_0 ?

$$a_n, n = 1, 2, 3, \dots$$

$$b_n, n = 1, 2, 3, \dots$$

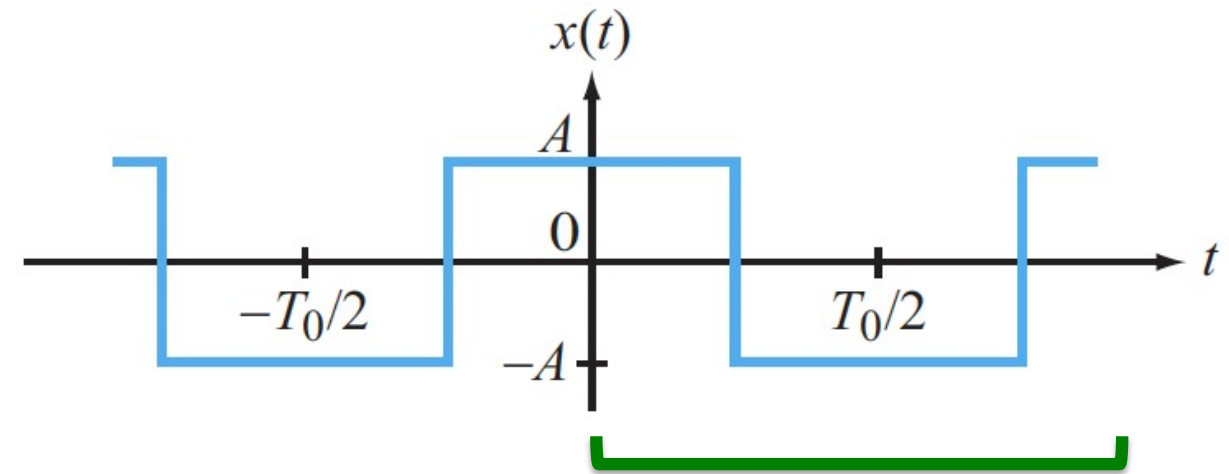


Example

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt = 0$$

$$a_n = \frac{2}{T_0} \int_0^{T_0} x(t) \cos(n\omega_0 t) dt, \quad n \geq 1$$

$$\begin{aligned} &= \frac{2}{T_0} \left[\int_0^{T_0/4} A \cos(n\omega_0 t) dt - \int_{T_0/4}^{3T_0/4} A \cos(n\omega_0 t) dt \right. \\ &\quad \left. + \int_{3T_0/4}^{T_0} A \cos(n\omega_0 t) dt \right] \end{aligned}$$





Recap: Integration Formula

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$



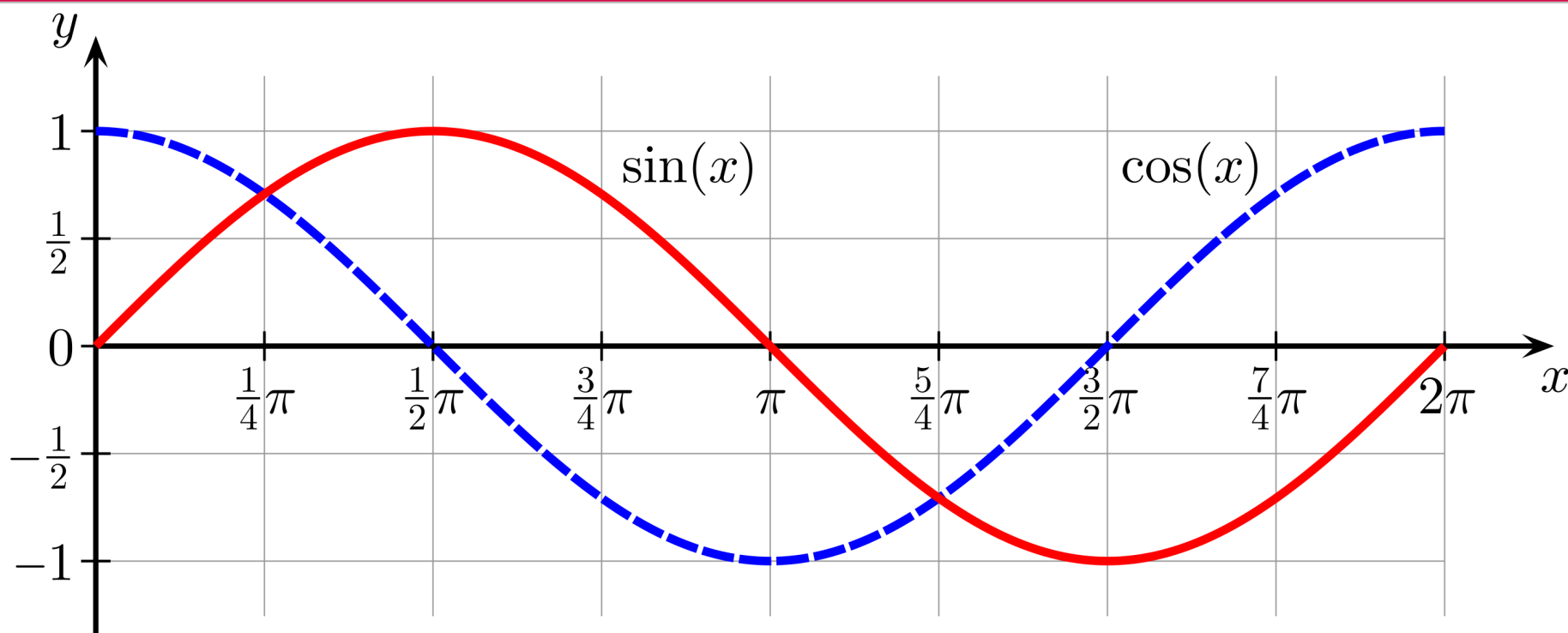
Example

$$\begin{aligned} a_n &= \frac{2A}{T_0} \left[\int_0^{T_0/4} \cos\left(\frac{n2\pi t}{T_0}\right) dt - \int_{T_0/4}^{3T_0/4} \cos\left(\frac{n2\pi t}{T_0}\right) dt + \int_{3T_0/4}^{T_0} \cos\left(\frac{n2\pi t}{T_0}\right) dt \right] \\ &= \frac{2A}{T_0} \left[\frac{T_0}{n2\pi} \sin\left(\frac{n2\pi t}{T_0}\right) \Big|_0^{T_0/4} - \frac{T_0}{n2\pi} \sin\left(\frac{n2\pi t}{T_0}\right) \Big|_{T_0/4}^{3T_0/4} + \frac{T_0}{n2\pi} \sin\left(\frac{n2\pi t}{T_0}\right) \Big|_{3T_0/4}^{T_0} \right] \\ &= \frac{A}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \left\{ \sin\left(\frac{3n\pi}{2}\right) - \sin\left(\frac{n\pi}{2}\right) \right\} + \left\{ \sin(2n\pi) - \sin\left(\frac{3n\pi}{2}\right) \right\} \right] \\ &= \frac{2A}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(\frac{3n\pi}{2}\right) \right] \end{aligned}$$

The answer depends upon n



Special Angles



$$\sin(n\pi) = 0, \text{ for } n = 0, 1, 2, 3, \dots$$

$$\cos(n\pi) = (-1)^n$$

$$\sin\left((2n-1)\frac{\pi}{2}\right) = (-1)^{n+1}$$

(ODD MULTIPLES OF $\pi/2$)

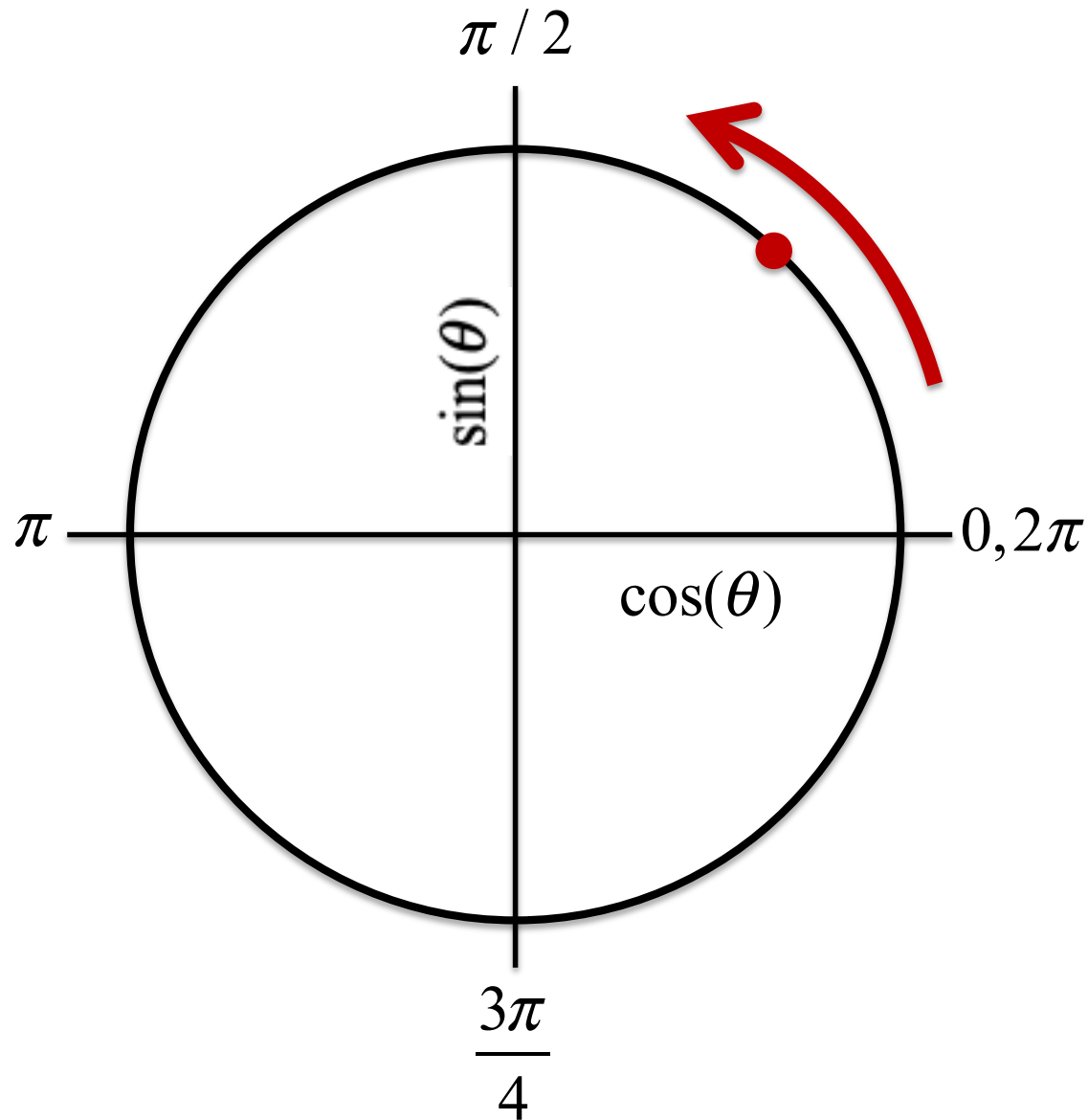
$$\cos(2n\pi) = 1$$

$$\cos((2n-1)\pi) = -1$$



Special Angles on the Unit Circle

READ OFF THE COSINE AND SINE VALUES FROM THE X AND Y AXES, RESPECTIVELY



$$\sin(n\pi) = 0, \text{ for } n = 0, 1, 2, 3, \dots$$

$$\sin\left(\frac{n\pi}{2}\right) = 0, 1, 0, -1, 0, \dots$$

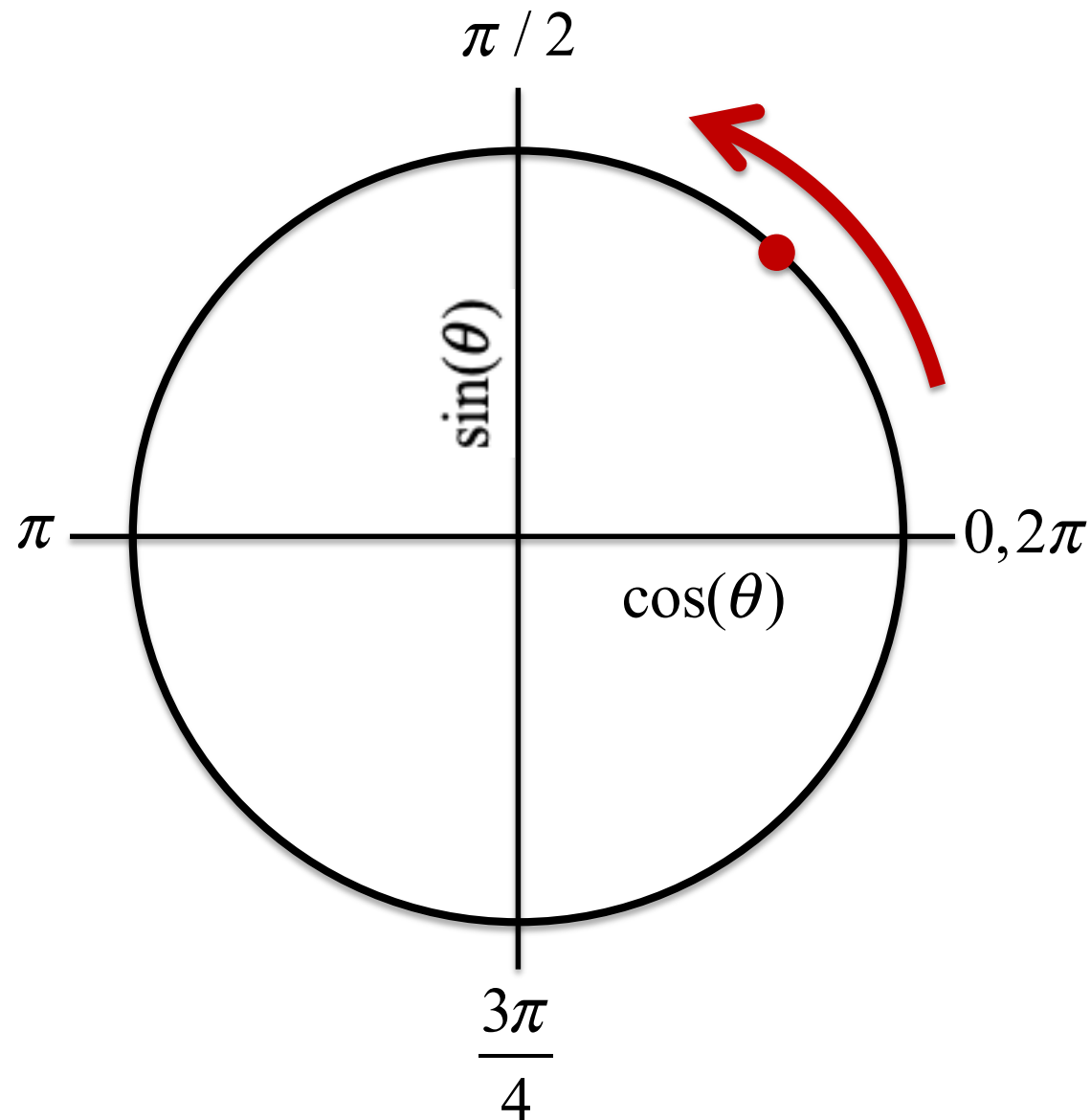
$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & \text{for even } n \\ (-1)^{n+1} & \text{for odd } n \end{cases}$$

$$\sin\left((2n-1)\frac{\pi}{2}\right) = (-1)^{n+1}$$



Special Angles on the Unit Circle

READ OFF THE COSINE AND SINE VALUES FROM THE X AND Y AXES, RESPECTIVELY



$$\cos(n\pi) = (-1)^n$$

$$\cos\left(\frac{n\pi}{2}\right) = 1, 0, -1, 0, 1, 0, \dots$$

$$\cos\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & \text{for odd } n \\ (-1)^n & \text{for even } n \end{cases}$$

$$\cos(2n\pi) = 1$$

$$\cos((2n-1)\pi) = -1$$



Example (Contd)

$$a_n = \frac{2A}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(\frac{3n\pi}{2}\right) \right]$$

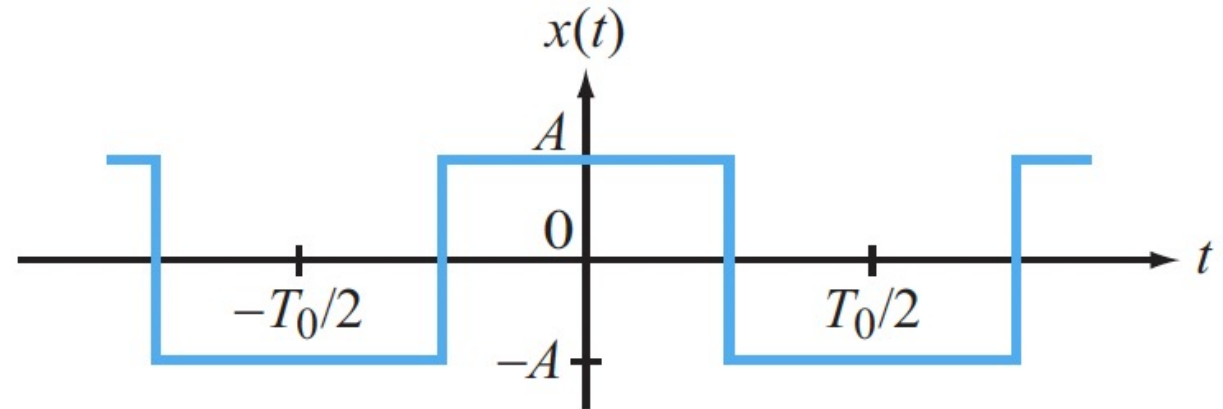
For $n = 1$,

$$a_1 = \frac{2A}{\pi} \left[\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{3\pi}{2}\right) \right] = \frac{2A}{\pi} [1 - (-1)] = \frac{4A}{\pi}$$

For $n = 2$,

$$a_2 = \frac{2A}{2\pi} \left[\sin\left(\frac{2\pi}{2}\right) - \sin\left(\frac{3 \times 2\pi}{2}\right) \right] = \frac{A}{\pi} [0 - 0] = 0$$

In fact, $a_n = 0$ for any even value of n





Example

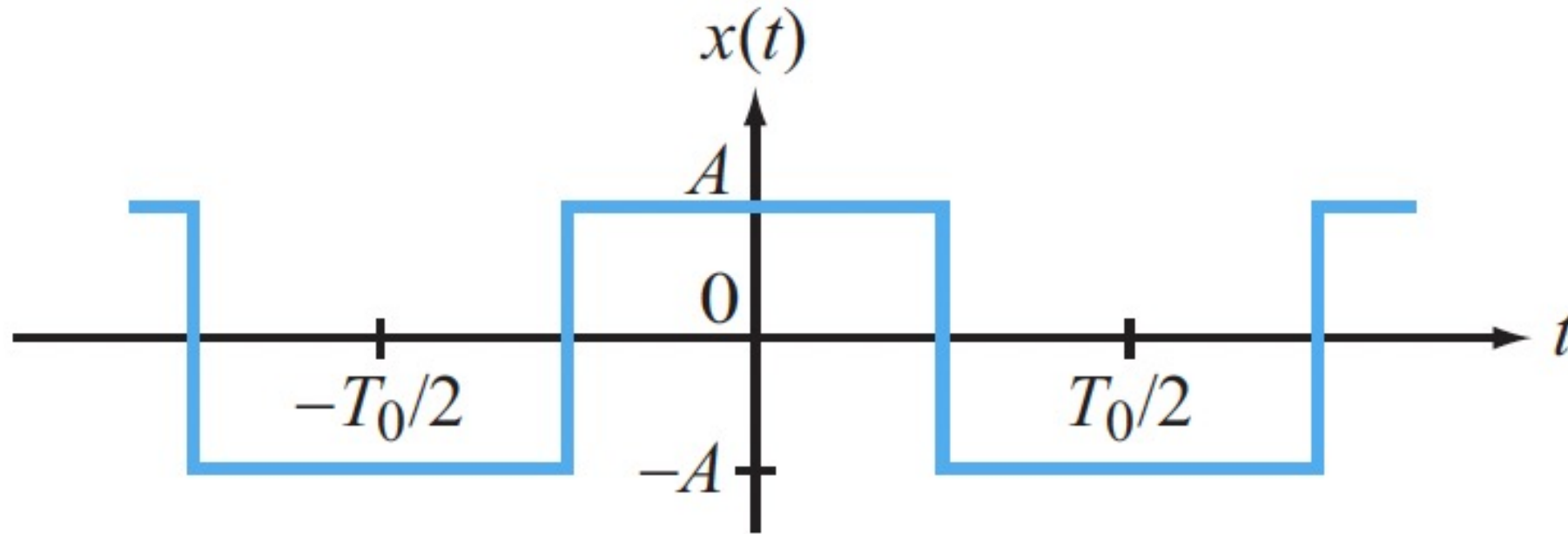
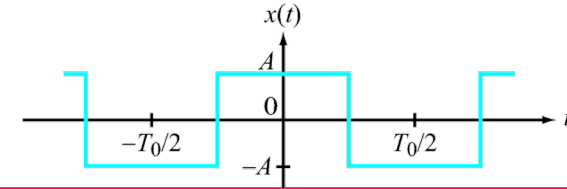
$$a_n = \frac{2A}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(\frac{3n\pi}{2}\right) \right]$$

For any odd value of n ,

$$a_n = \pm \frac{4A}{n\pi}$$



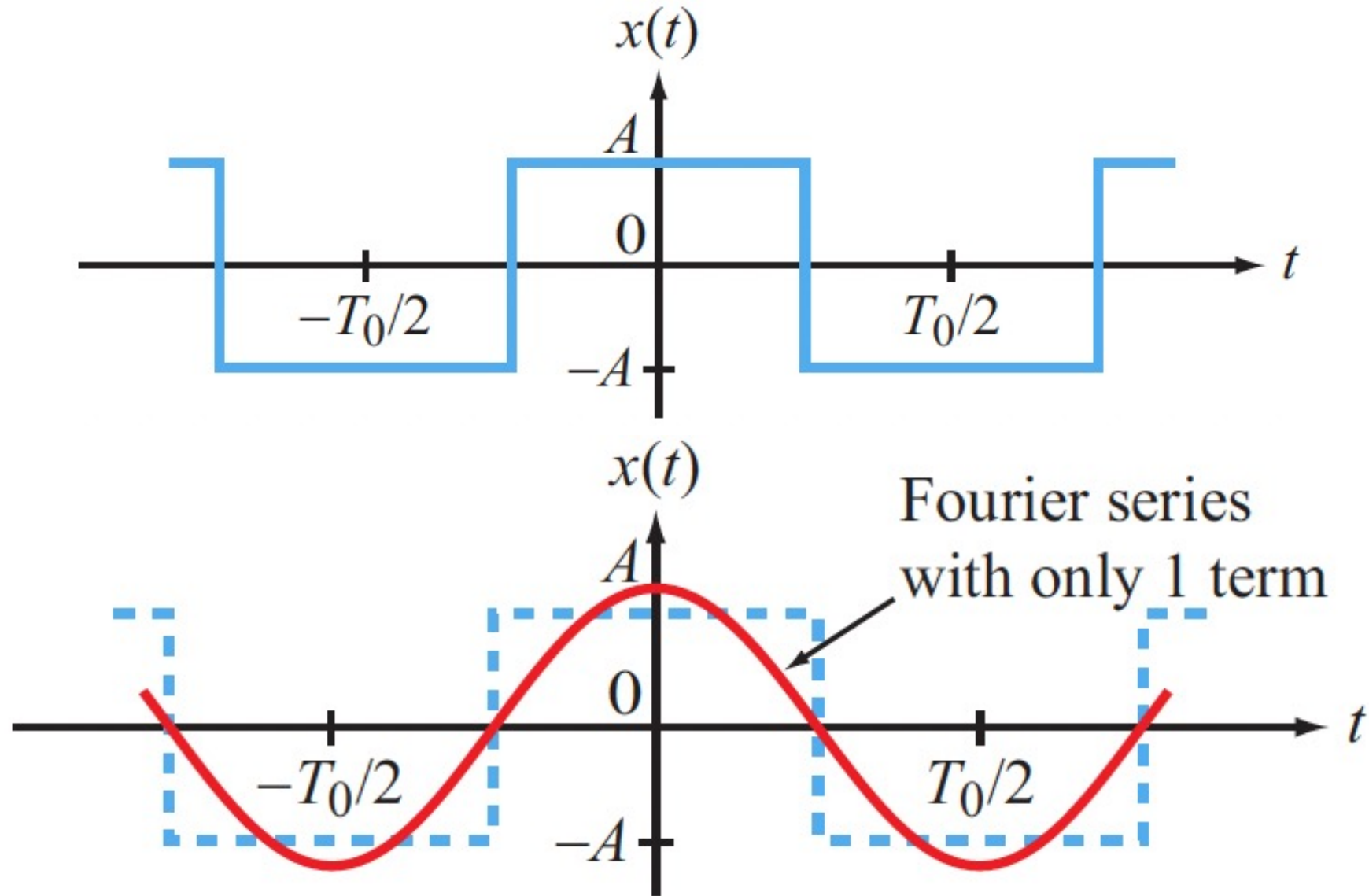
Example



$$x(t) = \frac{4A}{\pi} \cos\left(\frac{2\pi t}{T_0}\right) - \frac{4A}{3\pi} \cos\left(\frac{6\pi t}{T_0}\right) + \frac{4A}{5\pi} \cos\left(\frac{10\pi t}{T_0}\right) + \dots$$

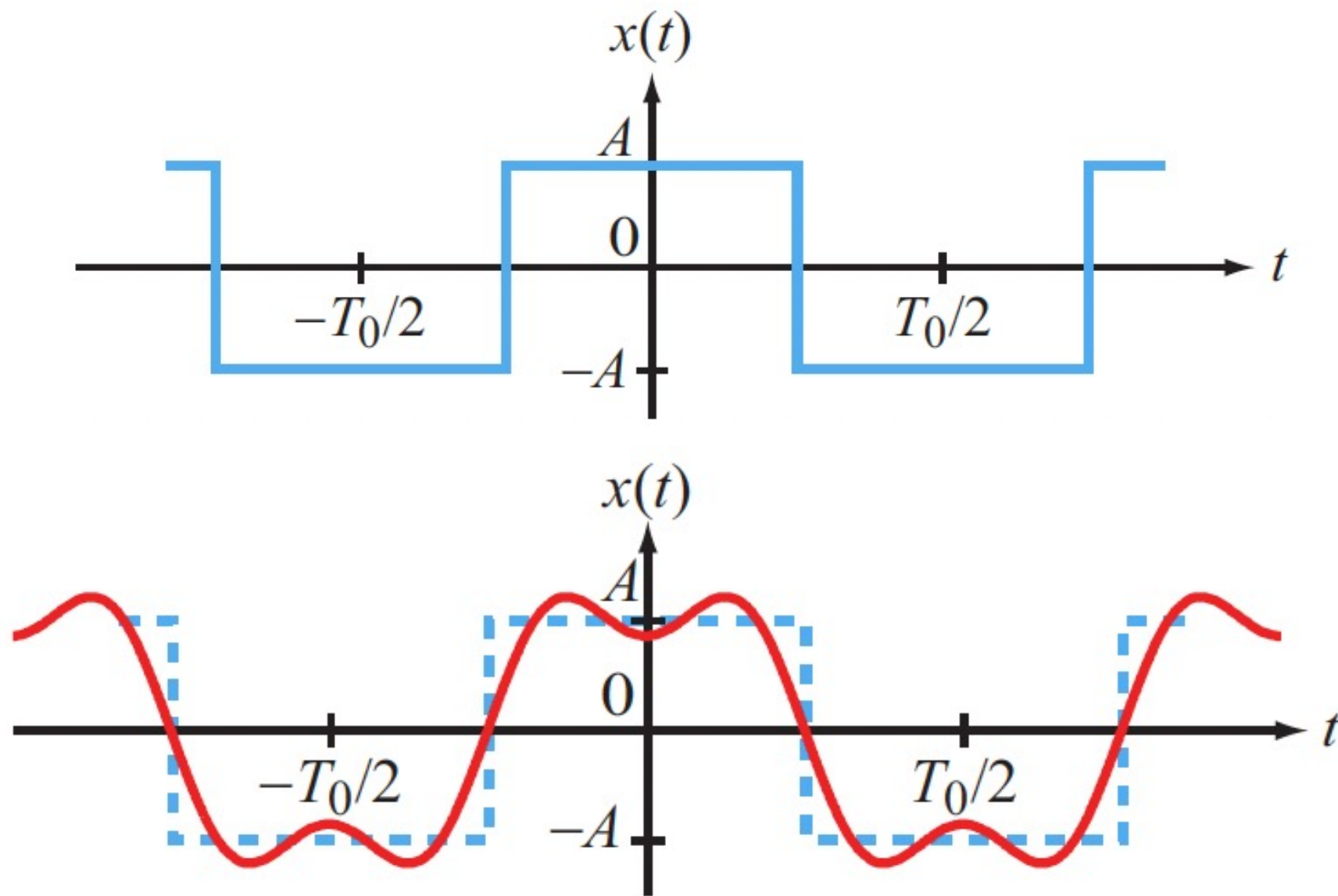


The First Term



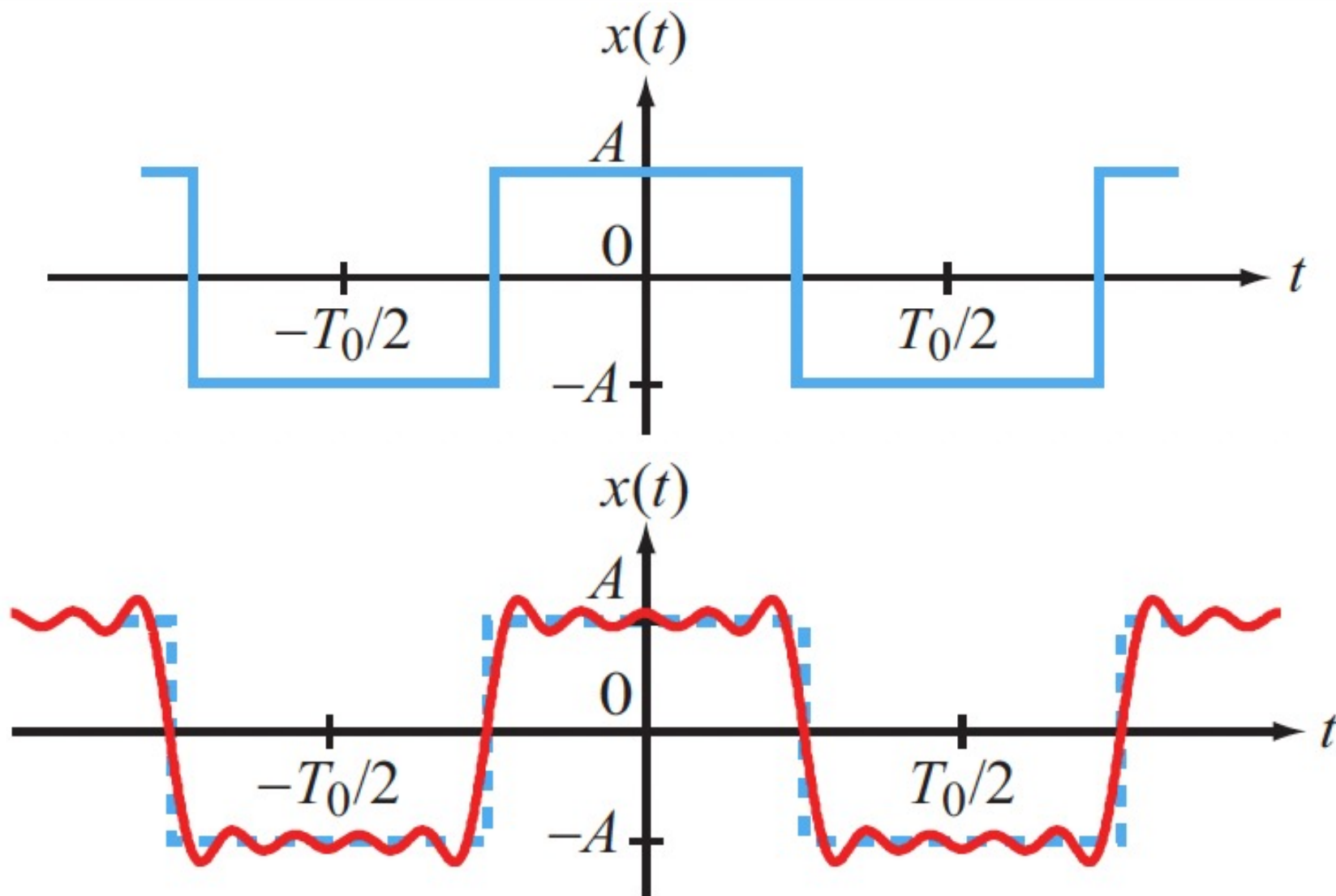


The First 3 Terms



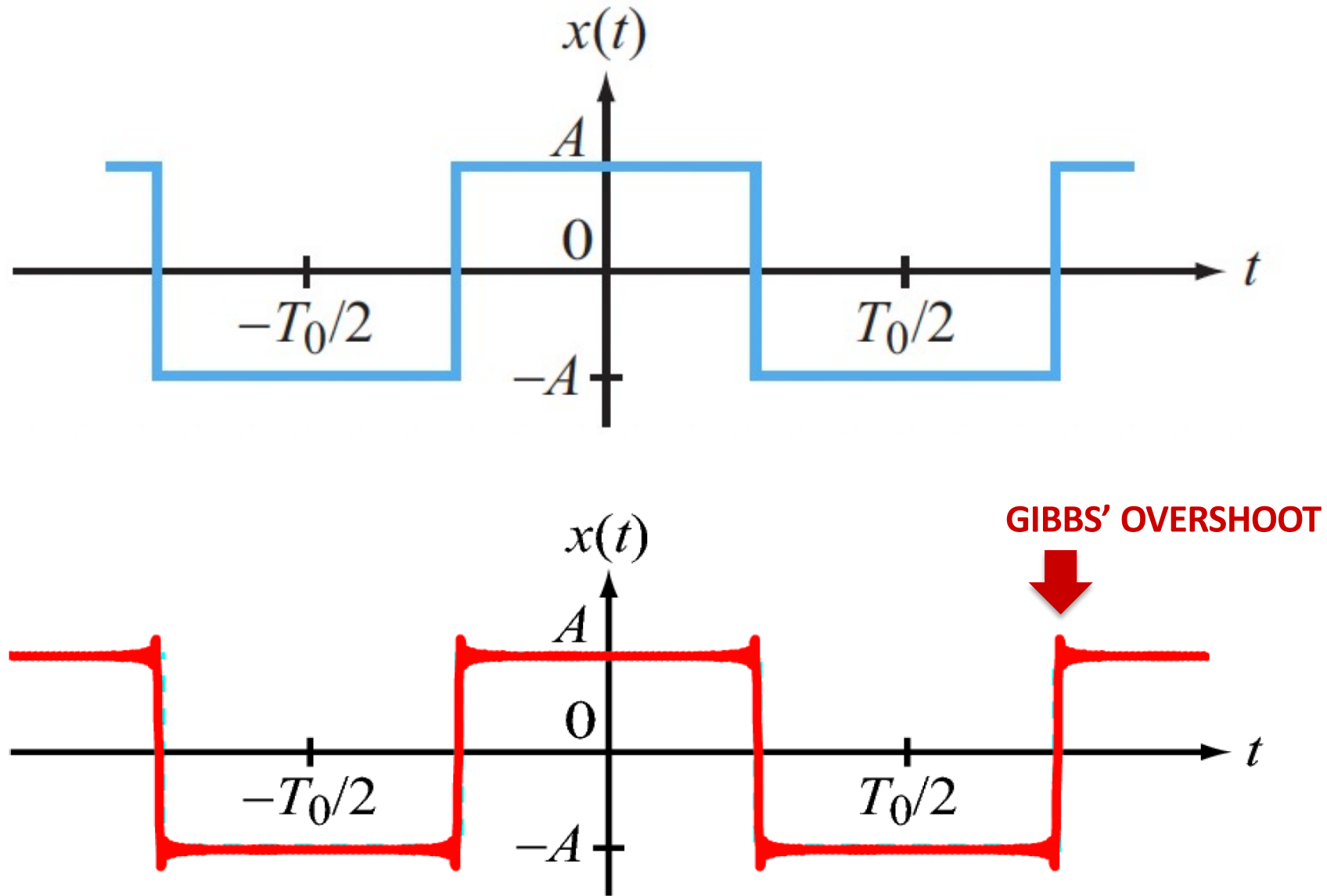


The First 10 Terms





The First 100 Terms: AWESOME

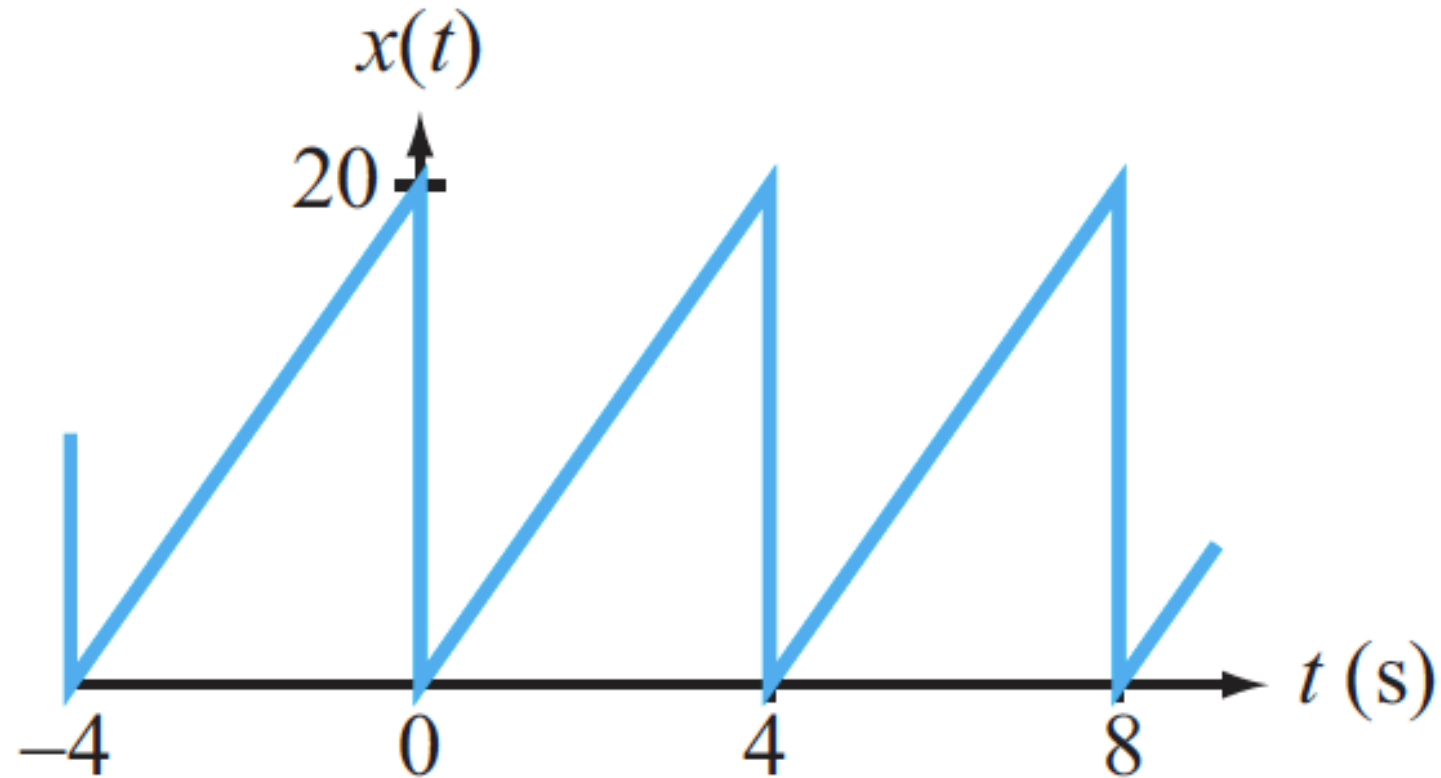


(e) Fourier series with 100 terms



Example: Sawtooth Waveform

$$T_0 = 4$$

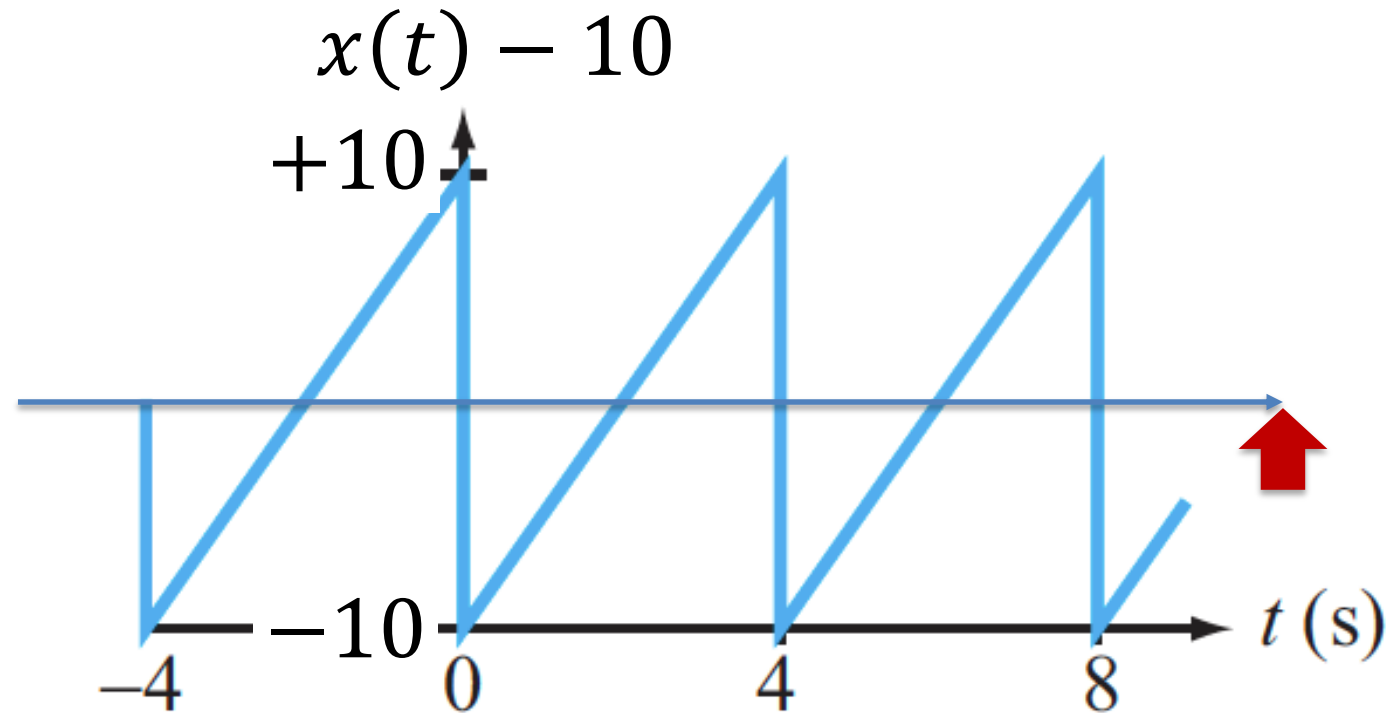


$$x(t) = 5t, \text{ when } 0 < t < 4$$

$$\Rightarrow a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt = \frac{1}{4} \int_0^4 5t dt = 10$$



Shift the axes to help reveal symmetries

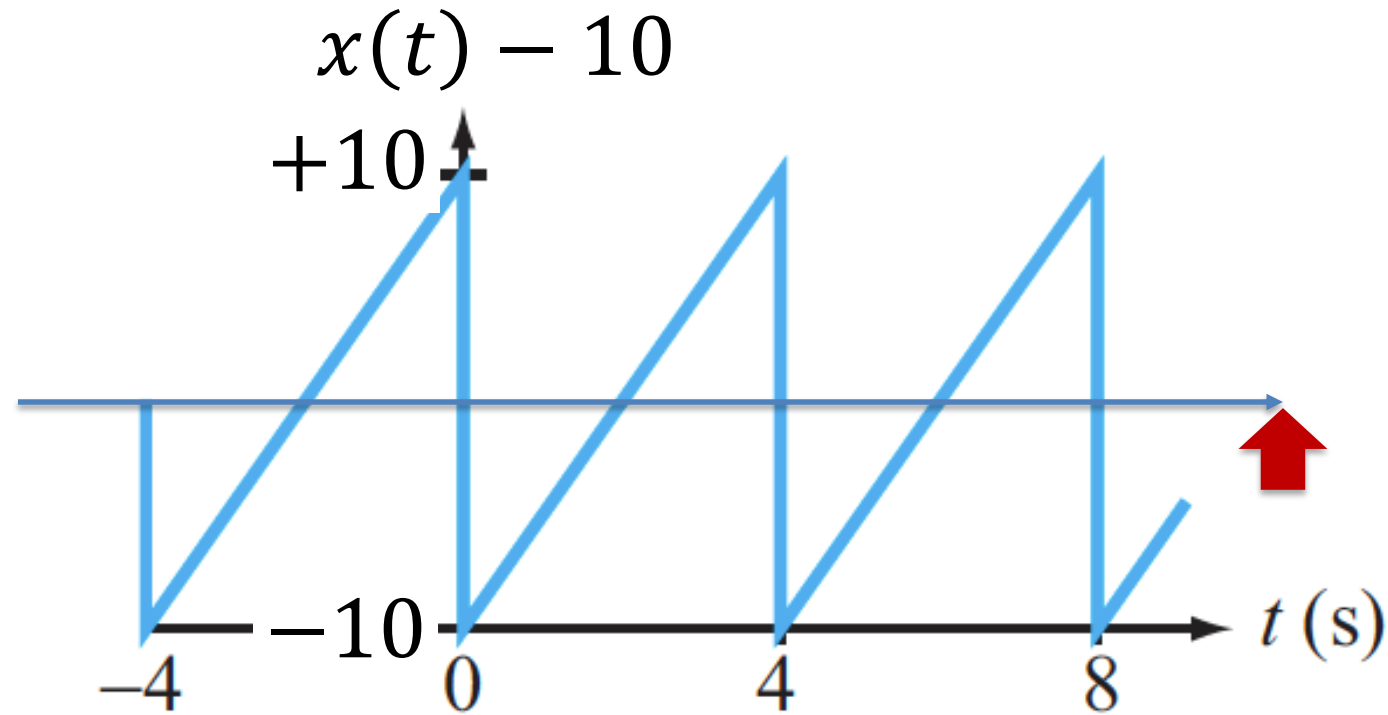


Subtracting the zero-frequency value reveals that the waveform has an odd symmetry

$$\Rightarrow a_n = \frac{2}{4} \int_0^4 5t \cos\left(\frac{n\pi}{2}t\right) dt = 0$$



Shift the axes to help reveal symmetries



We only need to calculate the b'_n s

$$b_n = \frac{2}{4} \int_0^4 5t \sin\left(\frac{n\pi}{2}t\right) dt = -\frac{20}{n\pi}$$



Activity

By performing integration by parts, show that:

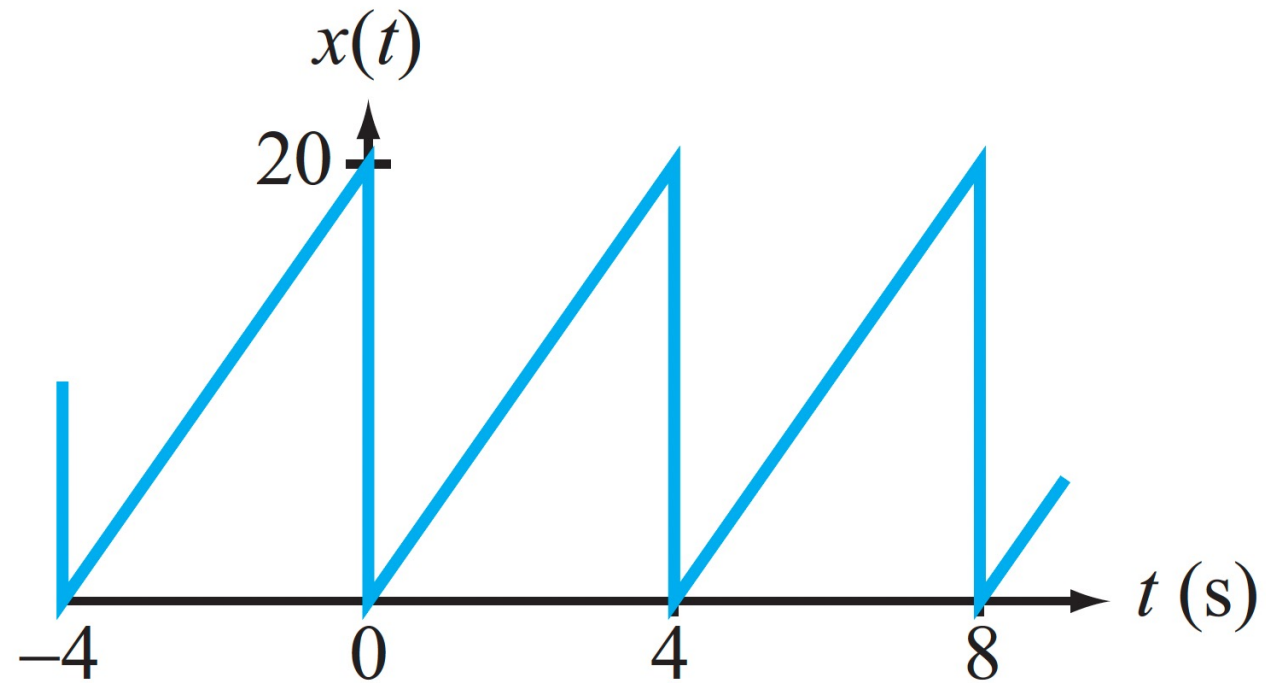
$$b_n = \frac{2}{4} \int_0^4 5t \sin\left(\frac{n\pi}{2}t\right) dt = -\frac{20}{n\pi}$$

$$x(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}t\right)$$



Progressive Approximations

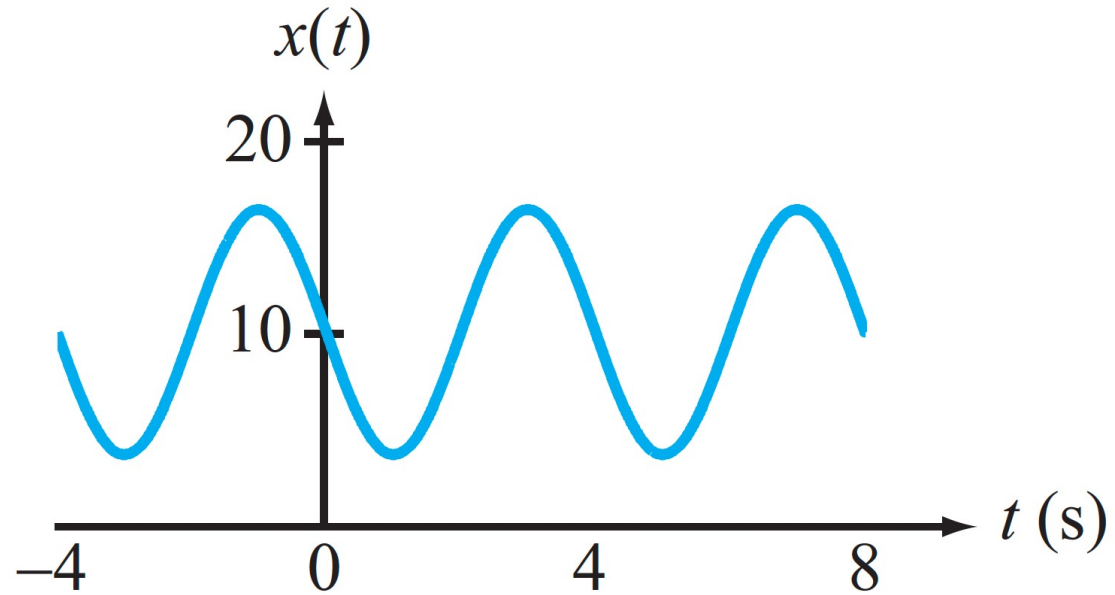
Original



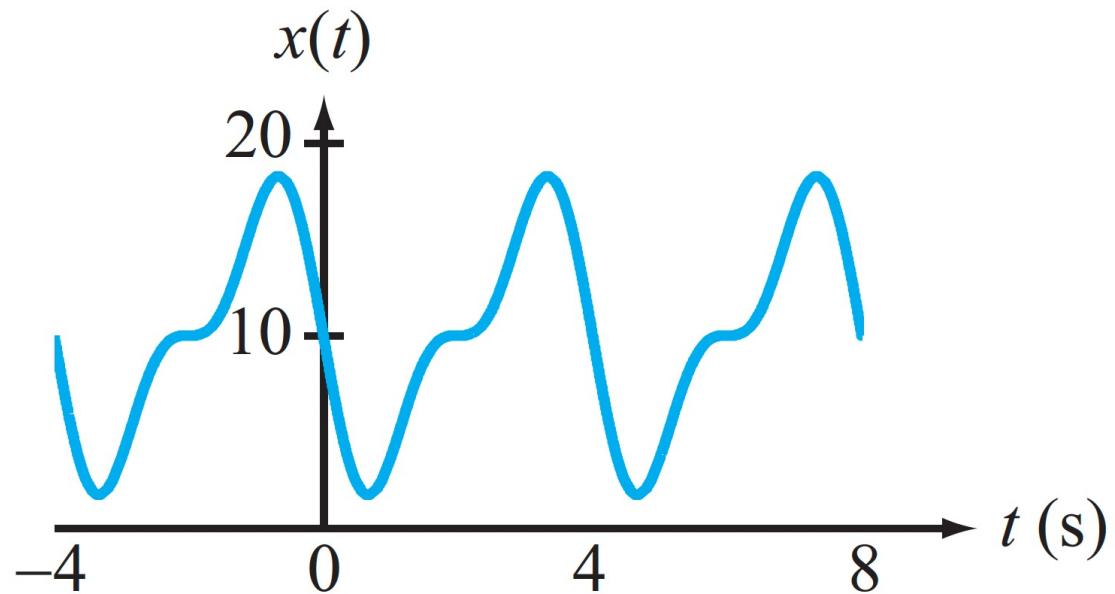


Progressive Approximations

$n_{\max} = 1$

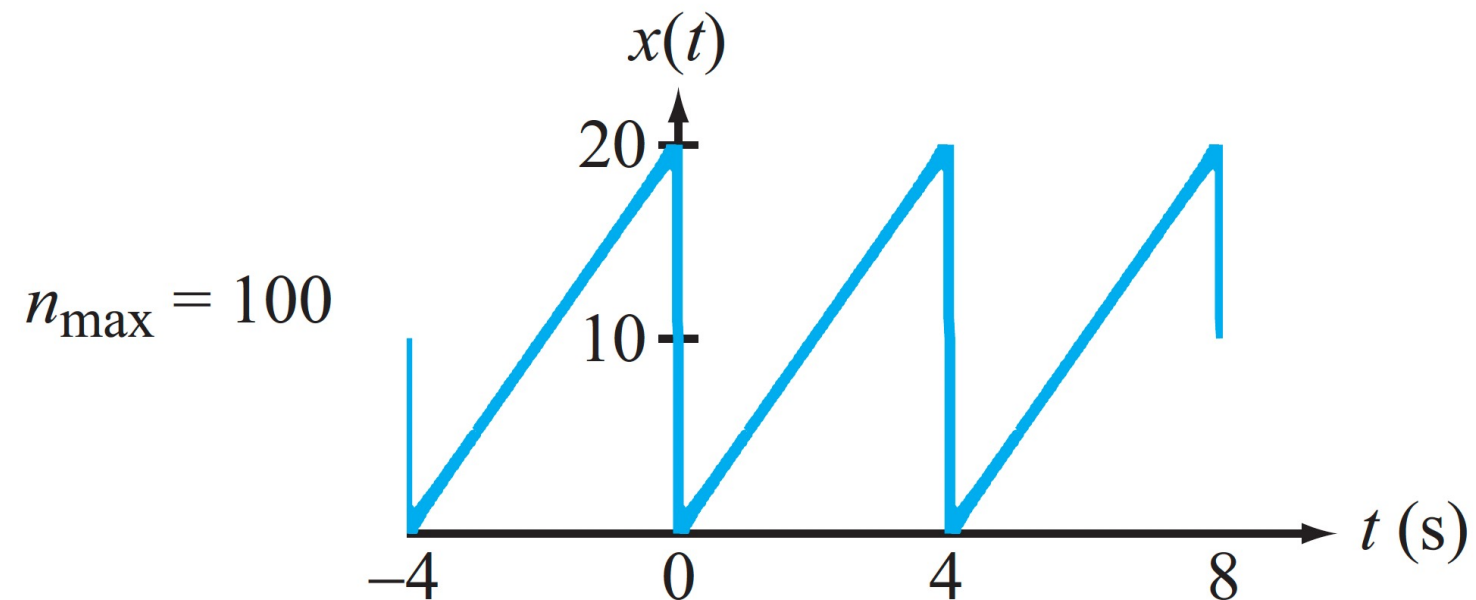
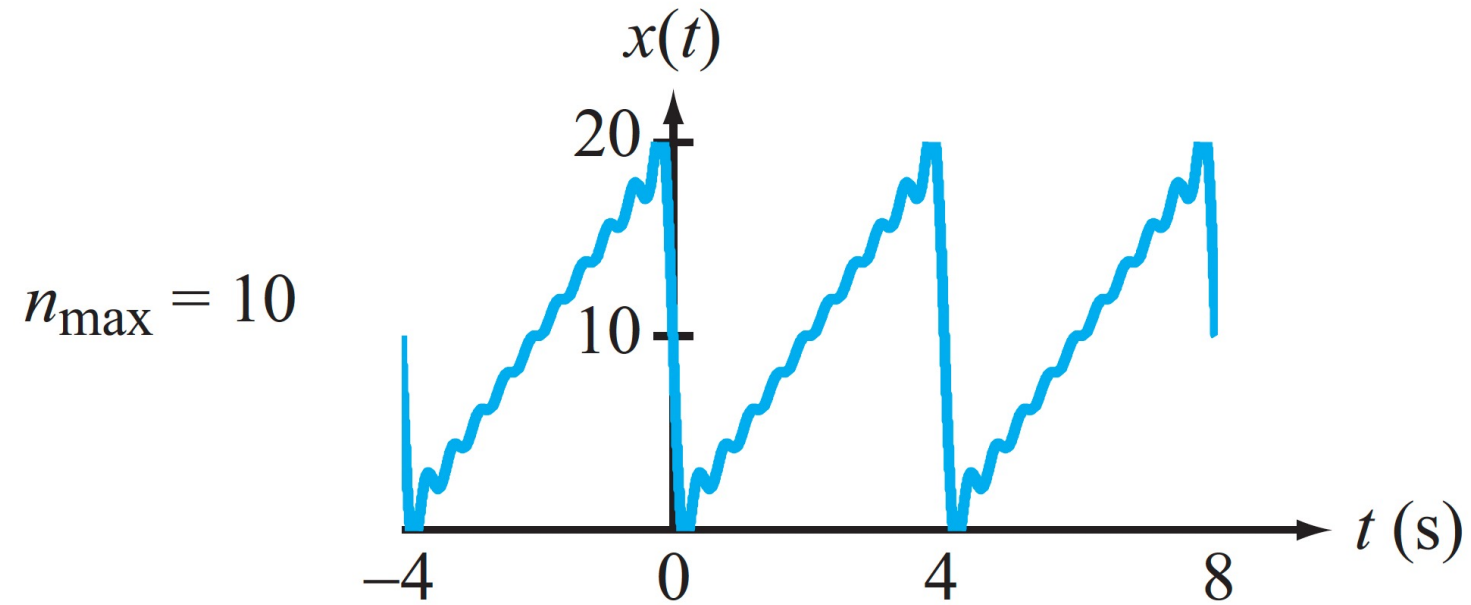


$n_{\max} = 2$





Progressive Approximations





Summary: Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)].$$

(sine/cosine representation) (5.27)

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt \quad a_n = \frac{2}{T_0} \int_0^{T_0} x(t) \cos(n\omega_0 t) dt$$
$$b_n = \frac{2}{T_0} \int_0^{T_0} x(t) \sin(n\omega_0 t) dt$$

A FUNDAMENTAL ENGINEERING TOOL



For the Next Class

- **Work through the textbook examples in Section 5-3 – 5-4.2**
- **Study Sections 5-4.3 – 5.5**

How to Study:

1. **Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**



Appendix: Useful Special Angles

Some of these cases occur often, so try to derive or remember them visually using the unit circle:

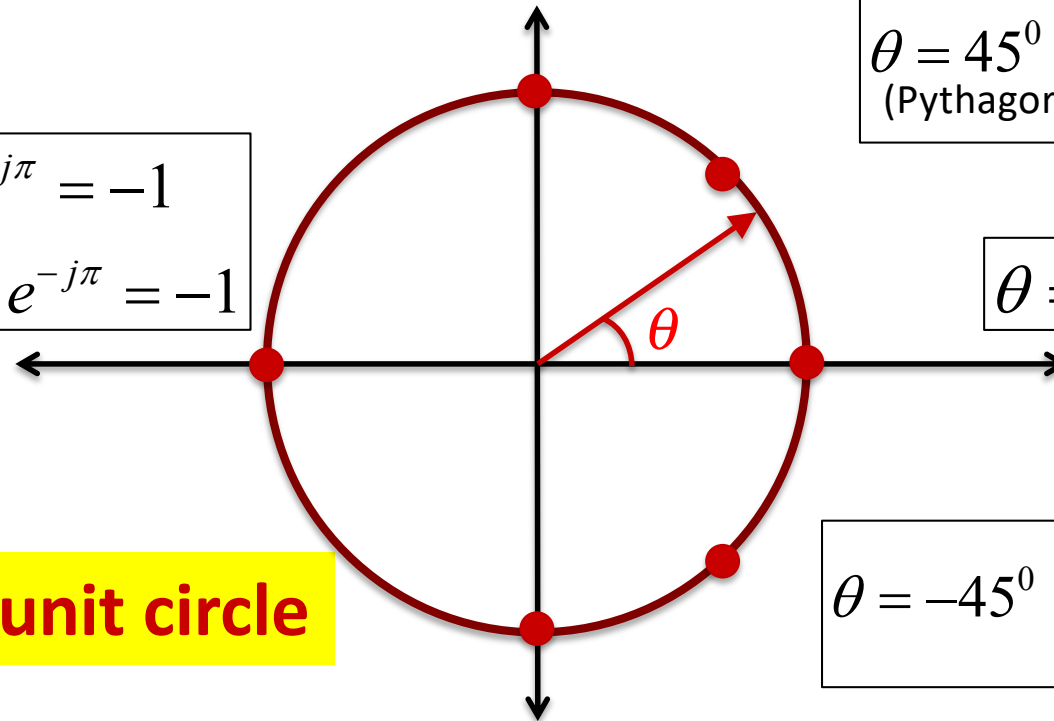
$$\theta = 90^\circ \rightarrow e^{j\pi/2} = \cos(90^\circ) + j \sin(90^\circ) = j$$

$$\theta = 45^\circ \rightarrow e^{j\pi/4} = \frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}} \quad (\text{Pythagoras' Theorem})$$

$$\begin{aligned} \theta = 180^\circ &\rightarrow e^{j\pi} = -1 \\ \theta = -180^\circ &\rightarrow e^{-j\pi} = -1 \end{aligned}$$

$$\theta = 0^\circ \rightarrow e^{j0} = 1$$

The unit circle



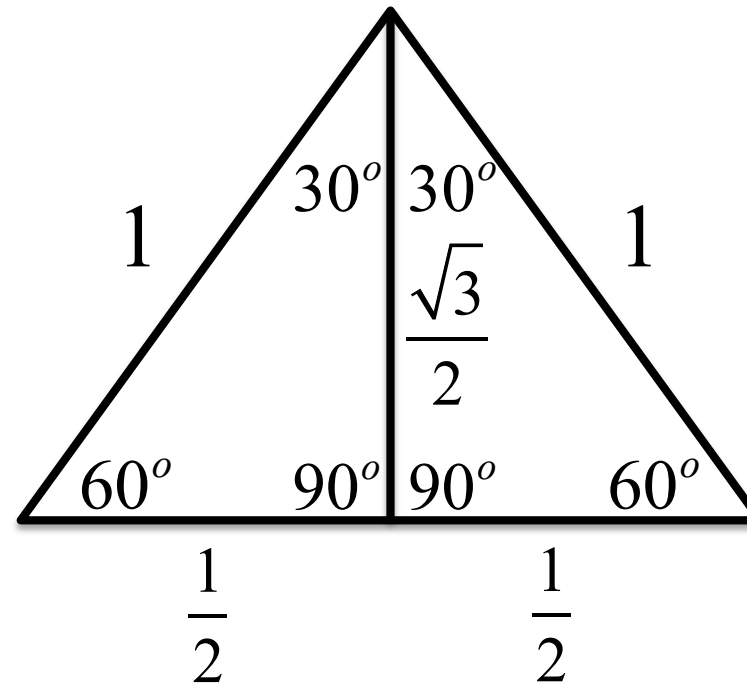
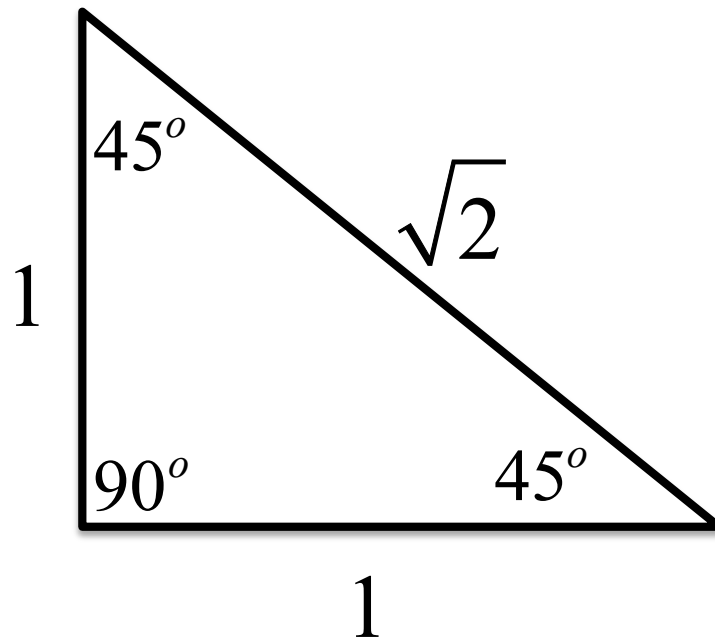
$$\theta = -45^\circ \rightarrow e^{-j\pi/4} = \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}}$$

$$\theta = -90^\circ \rightarrow e^{-j\pi/2} = -j$$



Appendix: Useful Special Angles

Some of the common angles can be derived quickly, or recalled from trigonometry:



EXAMPLE: $\sin(60^\circ) = \frac{\sqrt{3}}{2}$



Appendix: Useful Special Angles

